MILITARY SPECIFICATION

ADAPTER, RECOIL MECHANISM ARMAMENT SUBSYSTEM, HELICOPTER, 20MM: M97E1 AND M97E2

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings, and provides quality assurance provisions for the fabrication of parts, assembling and packaging for Adapter Recoil Mechanism for Armament Subsystem Helicopter, 20MM: M97E1 and M97E2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6868	-	Inspection Process, Magnetic Particle
MIL-W-63150	-	Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	-	Nondestructive Testing Personnel Qualification and Certification

FSC 1005

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT

11830567	-	Adapter, Recoil Mechanism

PACKAGING

11830567	-	Adapter, Recoil Mechanism (Packaging Data Sheet)

PUBLICATIONS

SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS

SQAP 7269485	-	Retainer, Spindle

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurment functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18	-	Method of Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B 117	-	Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Parts. All parts shall comply with applicable drawings, specifications and standards.

3.3 Spindle retainer. The Spindle Retainer shall comply with SQAP 7269485.

3.4 Recoil mechanism adapter. The Recoil Mechanism Adapter shall comply with all requirements specified on Drawing (Dwg.) D 11830567, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, cracks, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agents. Exterior surface coatings shall be continous; however, a few light scratches not exposing base material may be permitted. All required markings and stamping shall be neat and sharply defined. (Required packing shall be dry).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Spindle Assembly	D105D3854	10
Bushing, Floating	B162B8055	10
Bushing, FWD	B162B8066	10
Bushing, AFT	B162B8067	10
Ring, Inner	B162B8081	10
Ring, Outer	B162B8082	10
Ring, Inner, End	B162B8175	10
Ring, Outer, End	B162B8176	10
Spindle, Recoil, LH	C167C1558	10
Bearing, Sleeve	B196B7435	10
Housing, Recoil Adapter	F11830561	10
Housing Assembly, Recoil Adapter	C11830566	10
Adapter, Recoil Mechanism	D11830567	25

4.3.2 Inspections to be performed. See Table I specified herein.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen in the sample to comply with any of the stated requirements.

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TABLE I. FIRST ARTICLE INSPECTION.

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Adapter, Recoil Mechanism and Components				SEE BELOW
					NEXT HIGHER ASSEMBLY
					—
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spindle Assembly (Dwg. D105D3854				
	Examination for Defects	10		3.2	4.4.2.1
	Bushing, Floating (Dwg. B162B8055)				
	Examination for Defects	10		3.2	4.4.2.2
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Bushing, FWD (Dwg. B162B8056)				
	Examination for Defects	10		3.2	4.4.2.3
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Bushing, AFT (Dwg. B162B8057				
	Examination for Defects	10		3.2	4.4.2.4
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Ring, Inner (Dwg. B162B8081)				
	Examination for Defects	10		3.2	4.4.2.5
	Hardness	(a)		3.2	4.5.2

NOTES:

(a) This test shall be performed on the 10 samples subjected to the examination for defects.

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TABLE I. FIRST ARTICLE INSPECTION.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER SEE BELOW
Cont'd	Adapter, Recoil Mechanism and Components				NEXT HIGHER ASSEMBLY —
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ring, Outer (Dwg. B162B8082)				
	Examination for Defects	10		3.2	4.4.2.6
	Hardness	(a)		3.2	4.5.2
	Ring, Inner, End (Dwg. B162B8175)				
	Examination for Defects	10		3.2	4.4.2.7
	Hardness	(a)		3.2	4.5.2
	Ring, Outer, End (Dwg. B162B8176)				
	Examination for Defects	10		3.2	4.4.2.8
	Hardness	(a)		3.2	4.5.2
	Spindle, Recoil, LH (Dwg. C167C1558)				
	Examination for Defects	10		3.2	4.4.2.9
	Hardness	(a)		3.2	4.5.2
	Salt Spray Test	(a)		3.2	4.5.1
	Bearing Sleeve (Dwg. B196B7435)				
	Examination for Defects	10		3.2	4.4.2.10
	Hardness	(a)		3.2	4.5.2

NOTES:

(a) This test shall be performed on the 10 samples subjected to the examination for defects.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
Cont'd	Adapter, Recoil Mechanism and Components		3 OF 3	SEE BELOW
				NEXT HIGHER ASSEMBLY —

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Housing, Recoil Adapter (Dwg. F11830561)				
	Examination for Defects	10		3.2	4.4.2.11
	Hardness	(a)		3.2	4.5.2
	Housing Assembly, Recoil Adapter (Dwg. C11830566)				
	Examination for Defects	10		3.2	4.4.2.12
	Adapter, Recoil Mechanism (Dwg. D11830567)				
	Examination for Defects	10		3.2	4.4.2.13
	Functioning	(a)		3.2	4.5.5

NOTES:

(a) This test shall be performed on the 10 samples subjected to the examination for defects.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements.

4.4.2 Examination. The examinations listed in this specification for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans and procedures for major and minro defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER D10503854
4.4.2.1	Spindle Assembly (as disassembled from adapter, recoil mechanism)				NEXT HIGHER ASSEMBLY D11830567
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Free height "A"		0.65%	3.2	Gage
102	Free height "B"		0.65%	3.2	Gage
103	Free height "C"		0.65%	3.2	Gage
104	Free height "D"		0.65%	3.2	Gage
105	Deflection heights under specified loads improper (compression - extension characteristics)		0.65%	3.2	4.5.4
106	Part missing		0.65%	3.2	Visual
107	Part improperly assembled		0.65%	3.2	Visual
108	Part damaged (cracked, chipped edge, broken, dented)		0.65%	3.2	Visual
109	"Preloaded" information tag missing or marking incorrect or illegible (Note 2)		0.65%	3.2	Visual
Minor					
201	Identification marking misleading, illegible or missing		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Bushing, Floating		B162B8055 (sheets 1 and 2)
			NEXT HIGHER ASSEMBLY D105D3854

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	4.5.1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Total length		0.65%	3.2	Gage
104	Diameter of flange ("C")		0.65%	3.2	Gage
105	True position of flange diameter		0.65%	3.2	Gage
106	Diameter under flange ("B")		0.65%	3.2	Gage
107	True position diameter under flange		0.65%	3.2	Gage
108	Inside diameter		0.65%	3.2	Gage
109	Diameter of hole ("D")		0.65%	3.2	Gage
110	True position of hole		0.65%	3.2	Gage
111	Width of flange		0.65%	3.2	Gage
112	Magnetic particle inspection mark missing		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER B162B8055
4.4.2.2 cont'd	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY D105D3854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Thickness of through bottom		1.50%	3.2	Gage
202	Surface roughness (63)		1.50%	3.2	Visual - Measure
203	Part defective (crack, laps, seam, or other imperfections)		1.50%	3.2	Visual
204	Protective finish missing or defective		1.50%	3.2	Visual
205	Part number illegible, improper or missing		1.50%	3.2	Visual
206	Evidence of poor workmanship		2.50%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.3	Bushing, FWD		B162B8056 (sheets 1 and 2)
			NEXT HIGHER ASSEMBLY D105D3854

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Salt Spray test	4.5.1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Total length		0.65%	3.2	Gage
104	Diameter of flange ("C")		0.65%	3.2	Gage
105	True position of flange diameter		0.65%	3.2	Gage
106	Diameter under flange ("B")		0.65%	3.2	Gage
107	Inside diameter		0.65%	3.2	Gage
108	True position of inside diameter		0.65%	3.2	Gage
109	Width of flange		0.65%	3.2	Gage
110	Magnetic particle inspection mark missing		0.65%	3.2	Visual
111	Lubricant missing		0.65%	3.2	Visual
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual - Measure
202	Inside chamfer missing		1.50%	3.2	Visual
203	Part defective (crack, laps, seams, or other imperfections)		1.50%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3 cont'd	CONTINUATION SHEET				B162B8056
					NEXT HIGHER ASSEMBLY
					D10503854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
204	Protective finish missing or defective		1.50%	3.2	Visual
205	Part number illegible, improper or missing		1.50%	3.2	Visual
206	Evidence of poor workmanship		2.50%	3.6	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.4	Bushing, AFT		B162B8057 (sheets 1 and 2)
			NEXT HIGHER ASSEMBLY D105D3854

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	4.5.1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Total length		0.65%	3.2	Gage
104	Outside diameter		0.65%	3.2	Gage
105	True position of outside diameter		0.65%	3.2	Gage
106	Inside diameter		0.65%	3.2	Gage
107	Diameter of hole		0.65%	3.2	Gage
108	True position of hole		0.65%	3.2	Gage
109	Thickness of through bottom		0.65%	3.2	Gage
110	Magnetic particle inspection mark missing		0.65%	3.2	Visual
111	Lubricant missing		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4 cont'd	CONTINUATION SHEET				B16288057
					NEXT HIGHER ASSEMBLY
					D105D3854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual - Measure
202	Chamfer missing		1.50%	3.2	Visual
203	Part defective (crack, laps, seams, or other imperfections)		1.50%	3.2	Visual
204	Protective finish missing or defective		1.50%	3.2	Visual
205	Part number illegible, improper or missing		1.50%	3.2	Visual
206	Evidence of poor workmanship		2.50%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Ring, Inner		B162B8081
			NEXT HIGHER ASSEMBLY D105D3854

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Inside diameter ("A")		0.65%	3.2	Gage
103	Thickness ("D")		0.65%	3.2	Gage
104	Distance from top surface to basic diameter on tapered surface ("C") (2 places)		0.65%	3.2	Gage
105	Magnetic particle inspection mark missing		0.65%	3.2	Visual
106	Lubricant missing		0.65%	3.2	Visual
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual - Measure
202	Part defective (crack, laps, seams or other imperfections)		1.50%	3.2	Visual
203	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER B16288082
4.4.2.6	Ring, Outer				NEXT HIGHER ASSEMBLY D105D3854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Depth from bottom surface to inside basic diameter ("C") (2 places)		0.65%	3.2	Gage
103	Thickness ("D")		0.65%	3.2	Gage
104	Outside diameter ("A")		0.65%	3.2	Gage
105	Magnetic particle inspection mark missing		0.65%	3.2	Visual
106	Lubricant missing		0.65%	3.2	Visual
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual - Measure
202	Part defective (crack, laps, seams, or other imperfections)		1.50%	3.2	Visual
203	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Ring, Inner, End		B162B8175
			NEXT HIGHER ASSEMBLY
			D105D3854

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Inside diameter		0.65%	3.2	Gage
103	Thickness		0.65%	3.2	Gage
104	Distance between bottom surface and basic diameter on tapered surface ("C")		0.65%	3.2	Gage
105	Magnetic particle inspection mark missing		0.65%	3.2	Visual
106	Lubricant missing		0.65%	3.2	Visual
Minor					
201	Part defective (crack, laps, seams, or other imperfections)		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER B162B8176
4.4.2.8	Ring, Outer, End				NEXT HIGHER ASSEMBLY D105D3854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Outside diameter		0.65%	3.2	Gage
103	Depth from bottom surface to inside basic diameter("C")		0.65%	3.2	Gage
104	Thickness ("D")		0.65%	3.2	Gage
105	Magnetic particle inspection mark missing		0.65%	3.2	Visual
106	Lubricant missing		0.65%	3.2	Visual
Minor					
201	Part defective (crack, laps, seams, or other imperfections)		1.50%	3.2	Visual
202	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.9	Spindle, Recoil, LH				C167C1558
					NEXT HIGHER ASSEMBLY
					D105D3854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Salt Spray Test	4.5.1		3.2	4.5.1
102	Hardness		0.65%	3.2	4.5.2
103	Smaller outside diameter (.619)		0.65%	3.2	Gage
104	True position of smaller outside diameter		0.65%	3.2	Gage
105	Larger outside diameter		0.65%	3.2	Gage
106	Diameter of hole (spindle end)		0.65%	3.2	Gage
107	True position and perpendicularity of hole		0.65%	3.2	Gage
108	Location of hole (3.300)		0.65%	3.2	Gage
109	Length of shaft (7.88)		0.65%	3.2	Gage
110	Major diameter of thread		0.65%	3.2	Gage
111	Pitch diameter of thread		0.65%	3.2	Gage
112	Spherical radius at spindle end		0.65%	3.2	Gage
113	Length of thread		0.65%	3.2	Gage
114	Width of flat (.6225)		0.65%	3.2	Gage
115	True position of flat		0.65%	3.2	Gage
116	Projections on surface caused by hardness testing (where applicable)		0.65%	3.2	Visual
117	Magnetic particle inspection mark missing		0.65%	3.2	Visual
118	Lubricant missing		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER C167C1558
4.4.2.9 cont'd	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY D105D3854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual - Measure
202	Thread damaged		1.50%	3.2	Visual
203	Part defective (crack, laps, seams or other imperfections)		1.50%	3.2	Visual
204	Protective coating damaged		1.50%	3.2	Visual
205	Marking incorrect, illegible or missing		1.50%	3.2	Visual
206	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Bearing, Sleeve	1 OF 1	B196B7435
			NEXT HIGHER ASSEMBLY C11830566

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Magnetic particle		100%	3.2	4.5.3
Major					
101	Hardness		0.65%	3.2	4.5.2
102	Length		0.65%	3.2	Gage
103	Length to larger diameter		0.65%	3.2	Gage
104	Larger outside diameter		0.65%	3.2	Gage
105	True position larger outside diameter		0.65%	3.2	Gage
106	Smaller outside diameter		0.65%	3.2	Gage
107	True position of inside diameter		0.65%	3.2	Gage
108	Inside diameter		0.65%	3.2	Gage
109	Diameter across radius under flange (1.424)		0.65%	3.2	Gage
110	Magnetic particle inspection mark missing		0.65%	3.2	Visual
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual - Measure
202	Part defective (crack, laps, seams or other imperfections)		1.50%	3.2	Visual
203	Radius under flange missing		1.50%	3.2	Visual
204	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Housing, Recoil Adapter	1 OF 2	F11830561
			NEXT HIGHER ASSEMBLY C11830566

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of bearing cavity (1.4375)		0.65%	3.2	Gage
102	Diameter of spindle assembly ring cavity (1.769) (larger tolerance)		0.65%	3.2	Gage
103	Diameter of spindle assembly ring cavity for min depth (smaller tolerance)		0.65%	3.2	Gage
104	Depth to sleeve bearing seat (1.578)		0.65%	3.2	Gage
105	Distance between sleeve bearing seat and slot below spindle retainer cavity (6.997)		0.65%	3.2	Gage
106	Distance between datums surfaces -G- and -H-		0.65%	3.2	Gage
107	Distance between datum surface -G- and specified surface (.688)		0.65%	3.2	Gage
108	True position of sleeve bearing cavity		0.65%	3.2	Gage
109	True position of spindle assembly ring cavity		0.65%	3.2	Gage
110	Profile of arc (Zones 5-D, 7-D)		0.65%	3.2	Gage
111	Minor diameter of spindle retainer thread		0.65%	3.2	Gage
112	Pitch diameter of thread		0.65%	3.2	Gage
113	True position of countersink of thread		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.11 cont'd	CONTINUATION SHEET		2 OF 2	F11830561
				NEXT HIGHER ASSEMBLY
				C11830566

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter bolt holes (2) (spindle retainer end)		1.50%	3.2	Gage
202	Diameter bolt holes (2) (sleeve bearing end)		1.50%	3.2	Gage
203	True position bolt holes (2) (sleeve bearing end)		1.50%	3.2	Gage
204	Width of slot below spindle retainer thread		1.50%	3.2	Gage
205	Distance between bolt holes (2.40)		1.50%	3.2	Gage
206	Marking illegible, incorrect or missing		1.50%	3.2	Visual
207	Protective paint missing		1.50%	3.2	Visual
208	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63425(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.12	Housing Assembly, Recoil Adapter (as disassembled from Adapter, Recoil Mechanism)		1 OF 1		C11830566
					NEXT HIGHER ASSEMBLY D11830567
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of sleeve bearing		0.65%	3.2	Gage
102	True position of sleeve bearing		0.65%	3.2	Gage
103	Depth to sleeve bearing, max		0.65%	3.2	Gage
104	Lubricant missing		0.65%	3.2	Visual
Minor					
201	Surface roughness (63)		1.50%	3.2	Visual - Measure
202	Marking illegible, incorrect or missing		1.50%	3.2	Visual
203	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Adapter, Recoil Mechanism				D11830567
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Functioning		100%	3.2	4.5.5
102	Spindle retainer not finger tight		0.65%	3.2	Manual
103	Part or assembly damaged		0.65%	3.2	Visual
104	Part or assembly missing		0.65%	3.2	Visual
105	Part or assembly incorrectly assembled		0.65%	3.2	Visual
106	Grease missing from specified surfaces		0.65%	3.2	Visual
Minor					
201	Marking on instruction plate illegible, incorrect or missing		1.50%	3.2	Visual
202	Marking on identification plate illegible, incorrect or missing (Dwg. Notes 6 & 7)		1.50%	3.2	Visual
203	Marking on housing assembly illegible, incorrect or missing (Dwg. Note 4)		1.50%	3.2	Visual
204	Protective coatings missing (Dwg. Notes 5 & 6)		1.50%	3.2	Visual
205	Evidence of poor workmanship		2.50%	3.6	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (See 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Salt spray test. Five (5) of each specified part shall be selected from each lot and subjected to the salt spray test in accordance with the procedures specified in ASTM B 117. Unless otherwise specified on the applicable drawing the minimum exposure time shall be one and one half (1 1/2) hours using a 5% sodium chloride solution. If any part shows evidence of corrosion (red rust, salt deposits, etc.), it shall be classed defective and the lot shall be rejected.

4.5.2 Hardness. Samples shall be selected from each heat treatment batch (see NOTE) and subjected to this test. If any part fails to comply with the requirement specified on the applicable drawing, the part shall be classed defective. The hardness test shall be performed in accordance with the procedure specified in ASTM E 18. Each heat treat batch shall remain segregated until completion of all hardness tests.

NOTE: A heat-treatment batch shall be defined as parts that have been heat-treated at the same time in the same furnace and quench bath for each of all phases of the entire heat-treatment process.

4.5.3 Magnetic particle test. The procedures and equipment for performing this inspection shall be in accordance with MIL-I-6868 and MIL-W-63150 and shall be satisfactory to the Contracting Officer. Personnel performing this inspection shall be qualified in accordance with MIL-STD-410.

4.5.4 Deflection heights under load. The recoil mechanism adapter shall be tested for compression and extension characteristics by measuring the deflection heights under various loads using the procedures specified on Dwg. D105D3854. Any adapter which fails to comply with the specified compression or extension requirements shall be classed defective. Test equipment and procedures shall be satisfactory to the Contracting Officer.

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4.5.5 Functioning. The recoil mechanism adapter shall be subjected to the functioning test using the procedures specified on Dwg. D11830567. Any adapter which fails to comply with the specified functioning requirement or evidences degradation of its operational function shall be classed defective. Test equipment and procedures shall be satisfactory to the Contracting Officer.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. The recoil mechanism adapters shall be prepared, packaged, packed and marked in accordance with Packaging Data Sheet (PDS) 11830567 for the level of protection specified in the contract. (See 6.2)

6. NOTES

6.1 Intended use. The recoil mechanism adapter covered by this specification is intended for use with Gun, automatic, 20MM: M197 on Armament Subsystem, Helicopter, 20MM: M97E1 and M97E2.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

e. First Article sample requirements, if other than specified in Section 4 of this specification.

f. Packaging requirements, if other than specified in Section 5 of this specification.

g. Certificates of conformance for each lot or shipment of product.

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6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs (conforming to Data Item Description DI-E-1118 or DI-R-1714 with appropriate addendum to each). See 6.2

b. Data cards. When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Submission of inspection equipment design for approval. See MIL-W-63150. Equipment designs shall be submitted as required to: Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, NJ 07801. Request letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects and test paragraph.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAA-S.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

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Preparing Activity:
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Project 1005-A603

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-63425 Adapter, Recoil Mechanism Armament Subsystem, Helicopter, 20MM: M97E1 and M97E2

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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